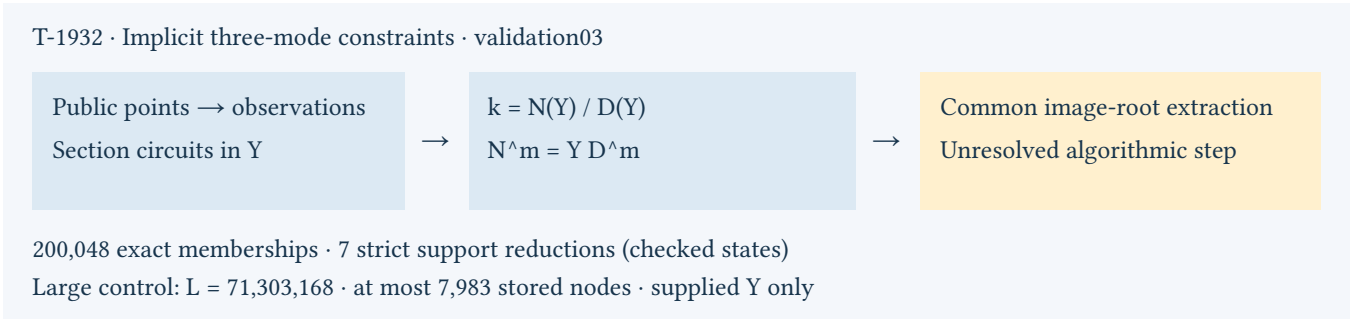
Implicit oriented-mode constraints

T-1932 · Prime fields · 2026-11-08

The builder retains exact three-mode image support before dense phase expansion. It adds absolute scalar consistency to the earlier relative constraints. It represents three Fourier modes at each supplied radius, not every individual sign.



Derived chart. Write (X−b)^h as the sum of X^e A(b,e)(Y), with X^m=Y. The observed modes are m A_0, mk A_1 and mk^(m−1) A(m−1). A nonzero plus mode gives k=fplus/(m A_1); otherwise a nonzero minus mode gives k=mY A(m−1)/fminus. Set k=N/D, impose N^m=YD^m, phase Y^L=epsilon, and all cross-multiplied mode equations. These imply a nonzero denominator and exact transcript equivalence. All-zero oriented modes use a separate unanchored case.

Recorded exact checks. 1,936 systems and200,048 image-membership equalities pass. Seven checked states have stricter support than the relative oracle. There are1,998 conditional phase checks;1,876 plus-anchor and60 unanchored systems. A synthetic empty-support case covers the minus-only branch; positive minus-only support is not demonstrated here. PCC-20261108-ac7576 is unsigned producer evidence.

Compact representation. Six controls over prime998244353 with m7 and L71,303,168 use at most7,983 stored nodes. Their Y values are supplied fixtures. A quadratic-modulus specialization agrees at both roots. No dense phase polynomial is built by the representation; no common-root extractor is supplied.

Public binding. Across300 public targets,277 produce transcripts and23 exit through verified identity/endpoints. All277 phase checks use fixture Y and verify the returned scalar against the point. API source calls total5,885, excluding setup/calibration;298 single-call guards pass. No unknown-image recovery count is claimed.

Custody and failures. Validation01 missed an inherited include. Validation02 failed a presumed-positive strictness assertion. Both are preserved. Validation03 reports actual counts after adding larger orbit-width controls; seven copied sources and13 compiler dependencies are hash-bound. Fresh overflow-checked build/check limits90/60 seconds. No timing comparison or cloud submission.

Next step and costs. Extract common image roots effectively from the implicit system. Charge public source setup/calls, point shifts, interning, allocation, circuit construction and specialization, scalar inversion and verification. Node and coefficient-slot caps are not peak-RAM bounds.

Sources. Predecessor signed-mode, oriented and packed modules; FFTW unnormalized DFT conventions. Finite-field orthogonality and the chart argument are derived explicitly in DERIVATION.md. KB unavailable. Independent signed review pending.